

Pricing & Revenue Optimization Diagnostic

Recommendation Memo — Retail Product Portfolio (9 categories, 52 SKUs, Jan 2017–Mar 2018)

Business Question

Margins across the product portfolio have been flat despite steady sales volume. This analysis identifies which product categories have room to adjust price without disproportionately hurting demand, using historical price-quantity data and competitor pricing benchmarks.

Approach

For each of the 9 product categories, price elasticity of demand was estimated using a log-log regression of monthly quantity sold on unit price, controlling for product-level fixed effects to isolate the pure price effect from differences in product popularity. Each category's average price was then compared against the average of three tracked competitor prices to flag mispricing relative to the market.

Key Finding

No category showed a statistically confident opportunity to raise prices — every category with a significant elasticity estimate was highly elastic (demand falls more than proportionally as price rises). The actionable opportunity instead lies in categories priced above competitors despite being elastic: correcting this mispricing recovers lost volume and revenue without a blanket price cut across the portfolio.

Recommendation by Category

Category	Price vs. Competitors	Elasticity	Recommendation
Watches & Gifts	+12.8%	-3.05 (elastic)	Reduce price toward competitor benchmark. Projected +21% revenue.
Furniture & Decor	-10.4%	-7.41 (very elastic)	Hold current pricing. Already below competitors; raising price would sharply reduce volume.
Computers & Accessories	-7.3%	-2.49 (elastic)	Hold current pricing for the same reason.
Bed, Bath & Table	+0.8%	-3.41 (elastic)	Pricing already near competitive benchmark. No action needed.
Garden Tools, Health & Beauty	+39% / +118%	Not significant	Insufficient statistical confidence to act. Recommend collecting more granular data before pricing changes.

Bottom line: Reprice Watches & Gifts down toward the competitor benchmark for a projected ~21% revenue lift in that category (~\$44K in the historical window analyzed). Hold pricing elsewhere; two categories (Garden Tools, Health & Beauty) need more granular data before any pricing decision.

Prepared using Python (pandas, statsmodels) for analysis and Power BI for visualization. Full dashboard and methodology available on request.